Assignment 6: Technical Analysis using R, Visualization Phase

Toronto Cakes Portfolio Dashboard (R Shiny)

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BDA400 – Data Science Tools and Techniques
CDI College – Data Analytics With Capstone
November 2026

Project Description

I run Toronto Cakes, a custom cake business that takes its orders through Instagram. This is the final stage of my technical analysis project and builds on Assignment 2, which created the Toronto Cakes Portfolio: META (Instagram's parent company), COST (Costco), HSY (Hershey), MDLZ (Mondelez), GIS (General Mills) and L.TO (Loblaw). In this stage I turned that work into an interactive R Shiny dashboard.

The dashboard fetches historical prices from Yahoo Finance with quantmod (with a CSV upload as a second data source and error handling for failed downloads and missing data). The user chooses the stock, date range, time frame (Daily, Weekly, Monthly) and chart type (Candlestick, Line, Area). Moving Averages (SMA or EMA) and Bollinger Bands are overlaid on the price chart, and RSI, MACD and Volume appear as panels underneath; each indicator can be switched on or off and its settings changed.

Four customizable trading rules are included: Moving Average Crossover, RSI Overbought/Oversold, MACD Signal Crossover and a Combined rule. The bars on the chart are annotated with the Buy and Sell signals, or with Buy / Sell / Hold on every bar. The app also shows summary cards, a downloadable signals table and a Portfolio tab comparing all six stocks. The R script, README and screenshots of the app running with live data are in the repository.

Repository Link

https://github.com/may0elmady-ship-it/TechnicalAnalysis/tree/main/Assignment6